Password Policy Validation (Case Study 29)

Problem Statement

Check whether a sample password has at least eight characters and contains a digit.

Objectives

Practise strings, len(), loops, and conditions.

Assumptions

This is a classroom example only; it does not provide complete real-world password security.

Algorithm

1. Read a password. 2. Check its length. 3. Check whether any character is a digit. 4. Print each failed rule or accept the
sample password.

Flowchart (Text Form)

START → Read password → Check length → Check digit → Print failed rules or valid → STOP.

Pseudocode

```
p
```

Python Source Code

```
password = input("Enter a sample password: ")
valid = True
if len(password) &lt; 8:
    print("Failed: use at least 8 characters")
    valid = False
has_digit = False
for ch in password:
    if ch.isdigit():
        has_digit = True
if not has_digit:
    print("Failed: include at least one digit")
    valid = False
if valid:
    print("Password follows these sample rules")
```

Test Case and Expected Output

Input: hello123
Expected output: Password follows these sample rules

Conclusion

The program validates two simple password rules and explains failures clearly.